

Sinkhorn Algorithm

1. $\mathbf{K} \leftarrow \exp\{-\mathbf{C}/\varepsilon\}$

2. Repeat until convergence:

$$\mathbf{u} \leftarrow \mathbf{a} \oslash \mathbf{K}\mathbf{v}, \mathbf{v} \leftarrow \mathbf{b} \oslash \mathbf{K}^T\mathbf{u}$$

Sinkhorn's Algorithm

- Optimal $\mathbf{u}, \mathbf{v}$ must satisfy:

$$\mathbf{a} = \pi \mathbf{1}_m = \text{diag}(\mathbf{u}) \mathbf{K} \text{diag}(\mathbf{v}) \mathbf{1}_m = \text{diag}(\mathbf{u}) [\mathbf{K} \mathbf{v}] = \mathbf{u} \odot \mathbf{K} \mathbf{v}$$

$$\mathbf{b} = \pi^\top \mathbf{1}_n = \text{diag}(\mathbf{v}) \mathbf{K}^\top \text{diag}(\mathbf{u}) \mathbf{1}_n = \text{diag}(\mathbf{v}) [\mathbf{K}^\top \mathbf{u}] = \mathbf{v} \odot \mathbf{K}^\top \mathbf{u}$$

- Iteratively compute:

$$\left. \begin{array}{l} \mathbf{u} \leftarrow \mathbf{a} \oslash \mathbf{K} \mathbf{v} \\ \mathbf{v} \leftarrow \mathbf{b} \oslash \mathbf{K}^\top \mathbf{u} \end{array} \right\} \rightarrow \text{Sinkhorn iterations}$$

- Theorem [Sinkhorn 1964] $(\mathbf{u}, \mathbf{v})$ converges

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